

Introduction to Elementary Particles – David Griffiths

Notes by Gene

2026

Contents

1 A Hitchhiker's Guide to Particles	1
1.1 Dramatis Personae	1
1.2 Particle Dynamics	3
1.3 Quantum Electrodynamics	3
1.4 Quantum Chromodynamics	4
1.5 Weak Interactions	5
1.5.1 Quarks and Leptons	5
1.5.2 Skewed Quark Generations	6
1.6 Conservation Laws	6

An updated set of notes to supersede my handwritten ones in 2025.

1 A Hitchhiker's Guide to Particles

1.1 Dramatis Personae

The Standard Model includes 17 subatomic particles (but not gravity).

1. Fermions (spin 1/2)
 - 6 Leptons: $e, \mu, \tau, \nu_e, \nu_\mu, \nu_\tau$ and antileptons
 - 6 Quarks: u, d, s, c, b, t and antiquarks
2. Gauge bosons (spin 1)
 - Gluon (strong force carrier): g
 - Photon (electromagnetic force carrier): γ
 - Intermediate vector bosons (weak force carriers): W^+, W^-, Z
3. Scalar bosons (spin 0)
 - Higgs boson (gives particles mass): H

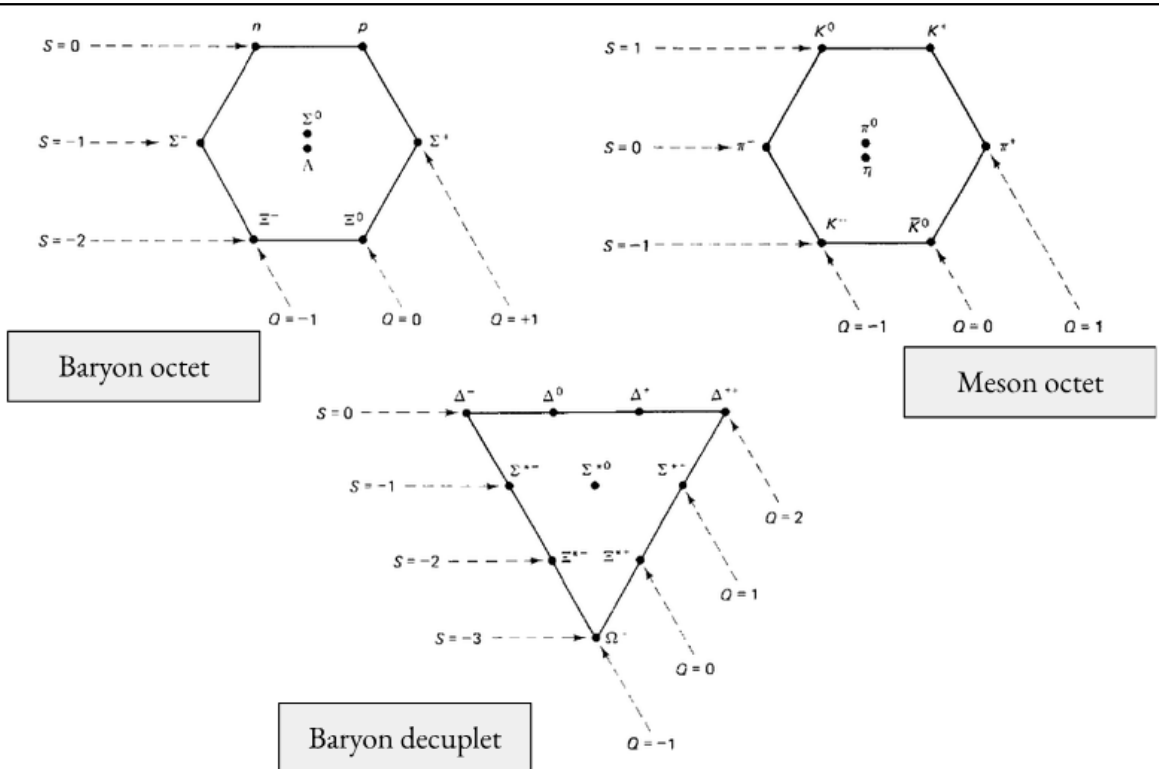
Lepton Families

	l	Q	L_e	L_μ	L_τ
1st generation	e	-1	1	0	0
	ν_e	0	1	0	0
2nd generation	μ	-1	0	1	0
	ν_μ	0	0	1	0
3rd generation	τ	-1	0	0	1
	ν_τ	0	0	0	1

Quark Families

	q	Q	D	U	S	C	B	T
1st generation	d	-1/3	-1	0	0	0	0	0
	u	2/3	0	1	0	0	0	0
2nd generation	s	-1/3	0	0	-1	0	0	0
	c	2/3	0	0	0	1	0	0
3rd generation	b	-1/3	0	0	0	0	-1	0
	t	2/3	0	0	0	0	0	1

Baryons and mesons can be organized into the baryon decuplet/octet and meson octet:



Baryon and meson octets/decuplets

1.2 Particle Dynamics

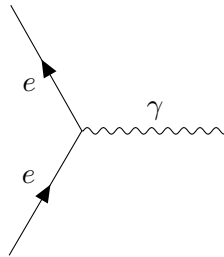
The Four Interactions			
Force	Strength	Theory	Mediator
Strong	10	Chromodynamics	Gluon
Electromagnetism	10^{-2}	Electrodynamics	Photon
Weak	10^{-13}	Flavordynamics	W, Z
Gravitational	10^{-42}	Geometrodynamics	Graviton

The mediators transmit forces between 1 lepton/quark to another.

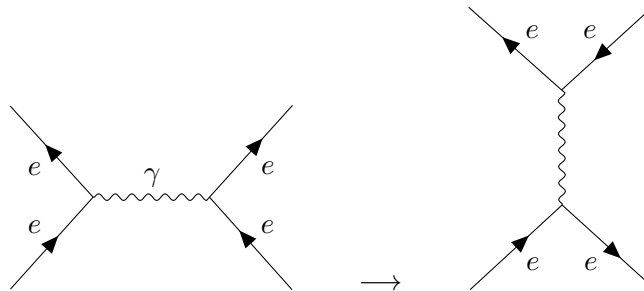
- Electromagnetism: formulated by Maxwell in 1860s.
- Quantum electrodynamics: formulated by Tomonaga, Feynman, Schwinger in 1940s.
- Weak interactions: formulated by Fermi, Lee & Yang, Feynman & Gell-Mann, etc. Until Glashow, Weinberg, Salam in 1960s.
 - GWS theory: weak & EM interactions are part of a unifying *electroweak interaction*.
- Strong interactions: formulated by Yukawa in 1930s, then others in 1970s.

1.3 Quantum Electrodynamics

All electromagnetic phenomena are reducible to the following Feynman diagram (read bottom to top):



I.e. e enters, emits/absorbs γ , and exits. The familiar Coulomb repulsion/attraction of charges (Møller scattering) is on the left. By rotating Møller scattering, we get Bhabha scattering (right): a particle-antiparticle pair annihilate, forming a photon, which later produces a particle-antiparticle pair.



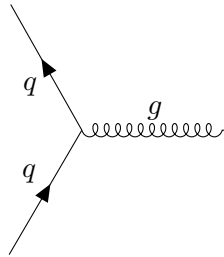
Note that only external lines describe observable particles. What goes on inside with virtual particles is allowed to violate conservation of mass, etc, as long as the external particles are all okay. Further,

the lines don't accurately represent trajectories. E.g. above, the interaction might be an attraction, not repulsion. All the diagram shows is that two particles go in, interact, and two particles come out.

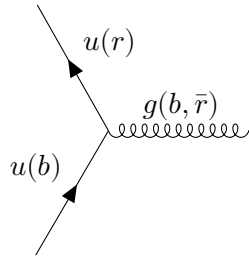
Each vertex introduces $\alpha = 1/137$, the coupling constant. So Feynman diagrams with many vertices, though they're physically possible, are negligible and we can sum up the possibilities.

1.4 Quantum Chromodynamics

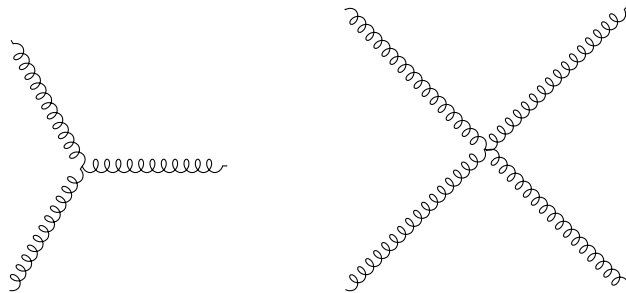
Color, not charge, causes the strong interaction. Gluons mediate the force between quarks. The fundamental vertex is



The flavor of the quark remains the same. However, color can change. E.g. blue up to red up quark, gluon carries away difference.



Gluons themselves have color (2), so primitive gluon vertices exist (only 3-gluon and 4-gluon vertices):



The coupling constant α_S depends on the separation distance of interacting particles.

- Quark polarization drives α_S up at short distances.
- Gluon polarization drives α_S down at short distances.

$$\text{Critical parameter: } a = 2f - 11n \quad \begin{cases} a < 0 & \text{QCD coupling decreases at short dist.} \\ a > 0 & \text{QCD coupling increases at short dist.} \end{cases}$$

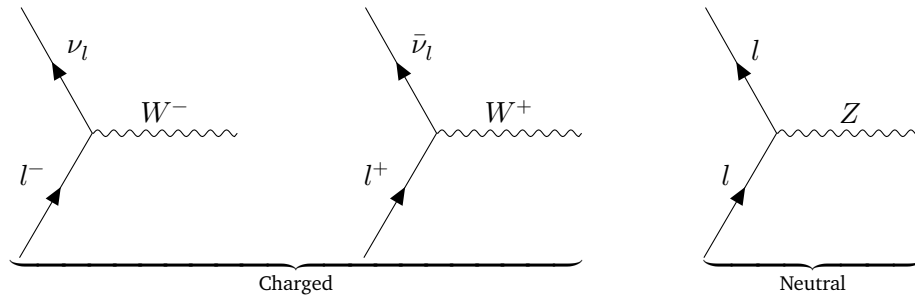
In the standard model ($f = 6$ flavors and $n = 3$ colors), $a < 0$ so QCD coupling decreases at short distances.

1.5 Weak Interactions

1.5.1 Quarks and Leptons

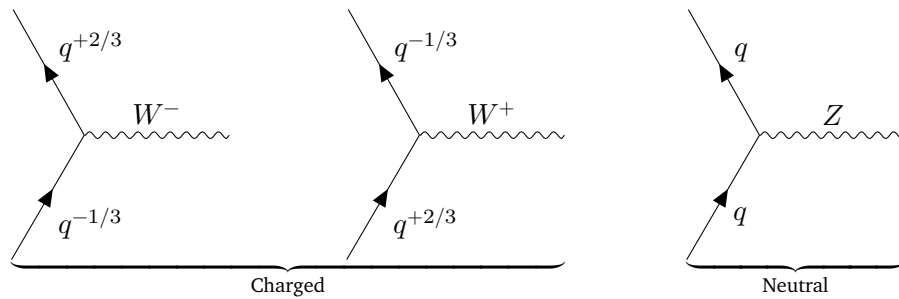
Quarks and leptons carry a “weak charge” (?) that leads to weak interactions. Weak interactions can be charged (mediated by $W^\pm$) or neutral (mediated by Z).

For leptons, the fundamental charged and neutral vertices are



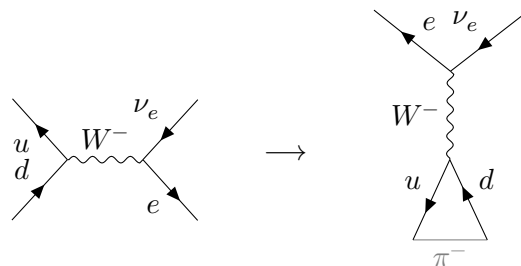
These vertices connect members to others of the **same generation**.

For quarks, same story: members of the same generation are connected (u, c, t to d, s, b , respectively). The fundamental charged and neutral vertices are

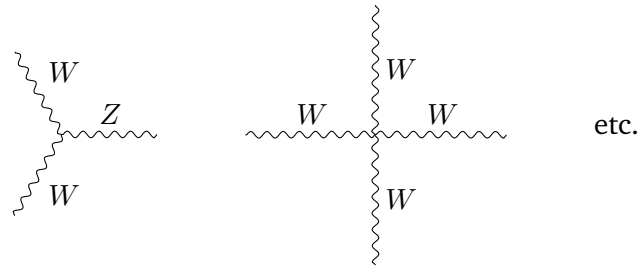


Color is conserved, but not flavor. I.e. **flavor is not conserved in weak interactions**.

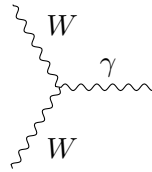
The weak interaction mediates decay. E.g. by rotating $d + \nu_e \rightarrow u + e$ (and enforcing quark confinement), we get $\pi^- \rightarrow e^- + \bar{\nu}_e$.



W and Z can also couple to each other:



Since W is charged, it can also couple to photons.



1.5.2 Skewed Quark Generations

The current available vertices don't explain strangeness-changing interactions, since the only interactions are between quarks of the same generation. Resolution: quark generations are skewed. They couple as

$$\begin{pmatrix} u \\ d' \end{pmatrix}, \begin{pmatrix} c \\ s' \end{pmatrix}, \begin{pmatrix} t \\ b' \end{pmatrix}$$

where

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \underbrace{\begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}}_{\text{CKM matrix}} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

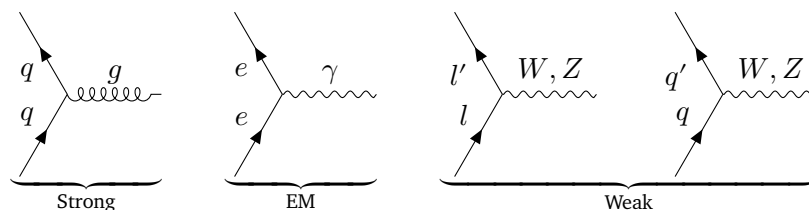
The CKM matrix is close to I , but there's some cross terms that explain strangeness-changing interactions.

1.6 Conservation Laws

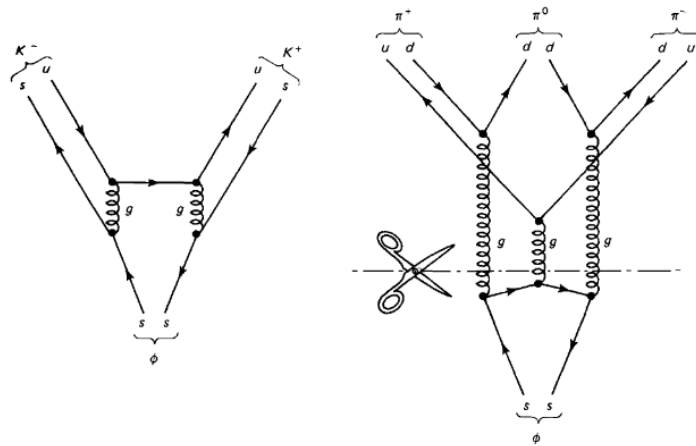
There are base kinematic conservation laws, from special relativity:

1. Conservation of energy & momentum
2. Conservation of angular momentum

There are also dynamic conservation laws for particle physics. Recall our fundamental vertices. Any quantity conserved in them must be conserved overall.



1. Conservation of charge
 2. Conservation of color
 3. Conservation of baryon number ($A = 1$ for baryons, $A = -1$ for antibaryons)
 4. Conservation of electron number, muon number, tau number
 5. Flavor is **approximately** conserved (except in weak interactions)
 6. OZI rule: if a Feynman diagram can be split in two by only cutting across gluon lines, the process is suppressed.
-



The OZI rule explains why $\phi \rightarrow K^- + K^+$ is more common than $\phi \rightarrow \pi^- + \pi^0 + \pi^+$.
